2016-2017 Virtual Challenge Meet #4 – Chemistry

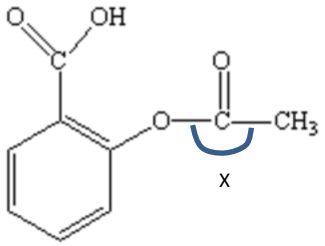
C11. In an electrochemical cell, if $E < E^o$ then
A) Q > 1
B) Q < 1
C) n = 1
D) the system is at equilibrium
E) K = 0

C12. Which of the following is likely to have the most positive entropy change?
A) $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3\ (g)$
B) $N_2(g) + 2\ O_2(g) \rightarrow 2\ NO_2\ (g)$
C) $2\ C(s) + O_2(g) \rightarrow 2\ CO(g)$
D) $C(s) + O_2(g) \rightarrow CO_2\ (g)$
E) $CaO(s) + CO_2(g) \rightarrow CaCO_3(s)$

C13. What is the solubility of silver nitrate in aqueous 0.25 M potassium chloride? (Ksp = 1.8×10^{-10})
A) 4.5×10^{-11} M
B) 7.2×10^{-10} M
C) 1.8×10^{-9} M
D) 1.7×10^{-7} M
E) 1.3×10^{-5} M

C14. Which of the following will have a pH of 3.0?
A) 0.001 M $HC_2H_3O_2$
B) 0.001 M NH_3
C) 0.001 M NH_4^+
D) 0.001 M HI
E) More than one answer choice is correct

C15. What is the approximate O-C-C bond angle noted by X in the molecule shown below?



A) 120°
B) 90°
C) 180°
D) 109°
E) unable to determine without more information

C16. A certain reaction occurring at constant pressure has ΔU = +45 kJ with 14 kJ of expansion work done by the system. What is the enthalpy change for this reaction?
A) -59 kJ
B) +45 kJ
C) -31 kJ
D) +31 kJ
E) +59 kJ

C17. When zinc metal is added to an aqueous solution containing nitrate ions and sodium hydroxide, ammonia and $Zn(OH)_4^{2-}$ are formed. What species is reduced?
A) nitrate
B) zinc metal
C) hydroxide
D) ammonia
E) water

C18. A titration required 20.3 mL of 0.600 M sodium hydroxide solution to completely neutralize 2.00 g of an unknown monoprotic binary acid. What is the molar mass of the acid?
A) 328 g/mol
B) 244 g/mol
C) 164 g/mol
D) 82.1 g/mol
E) 41.1 g/mol

C19. Consider the cell diagram
$Pt \mid Br^-(aq) \mid Br_2(l) \parallel Cl_2(g) \mid Cl^-(aq) \mid Pt$
Which reaction occurs at the cathode?
A) $Br_2(l) + 2\ e^- \rightarrow 2\ Br^-(aq)$
B) $2\ Br^-(aq) \rightarrow 2\ Br_2(l) + 2\ e^-$
C) $2\ Cl^-(aq) \rightarrow 2\ Cl_2(g) + 2\ e^-$
D) $Cl_2(g) + 2\ e^- \rightarrow 2\ Cl^-(aq)$
E) $Cl_2(g) + 2\ Br^-(aq) \rightarrow 2\ Cl^-(aq) + Br_2(l)$

C20. The standard enthalpy change for the decomposition of gaseous sulfur trioxide into gaseous sulfur dioxide and oxygen gas is +198 kJ. What conditions will favor production of sulfur dioxide in an equilibrium system for this reaction?
A) high temperature and low pressure
B) high temperature and high pressure
C) low temperature and high pressure
D) low temperature and low pressure
E) low temperature and small container volume